

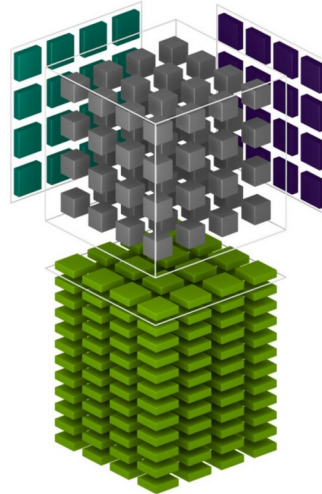


Module 0

Matrix / Tensor Algebra

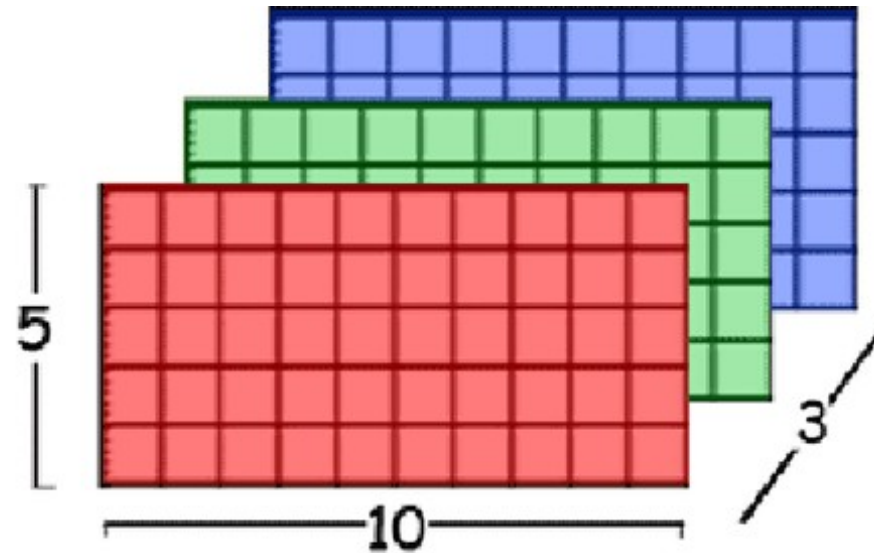
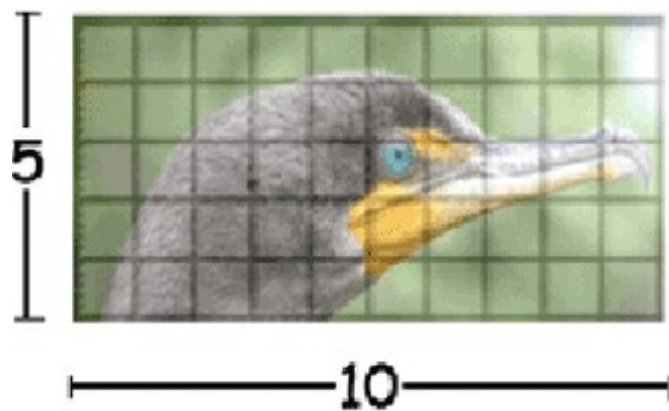


Matrices / Tensors



Data $\longrightarrow$ Vectors, Matrices, Tensors

- ❑ Feature creation in ML involve transforming raw data into numerical representations (vectors, matrices, tensors)
- ❑ For example, Images



Data → Vectors, Matrices, Tensors

□ Text, Documents

Documents

However, complexity.
We will see how small.
Given a function based.
Using entropy of traffic.
We study the complexity of influencing elections through bribery: How computationally complex is it for an external actor to determine whether by a certain amount of bribing voters a specified candidate can be made the election's winner? We study this problem for election systems as varied as scoring ...

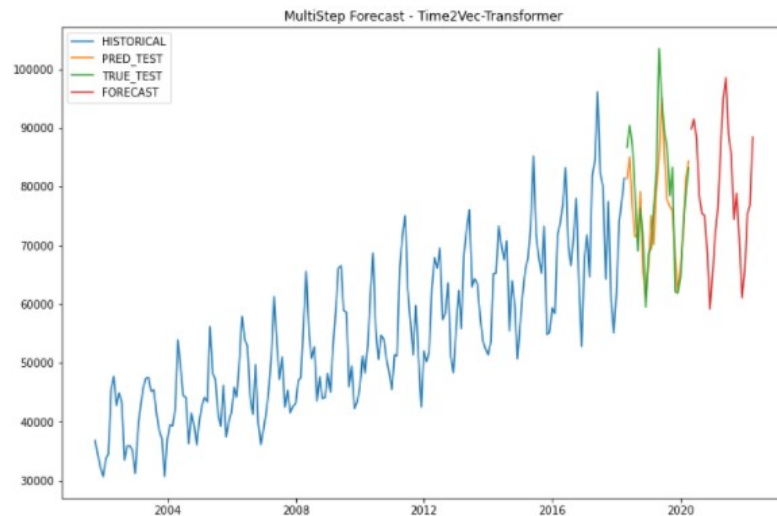
Vector-space representation

	D1	D2	D3	D4	D5
complexity	2		3	2	3
algorithm	3			4	4
entropy	1			2	
traffic		2	3		
network		1	4		

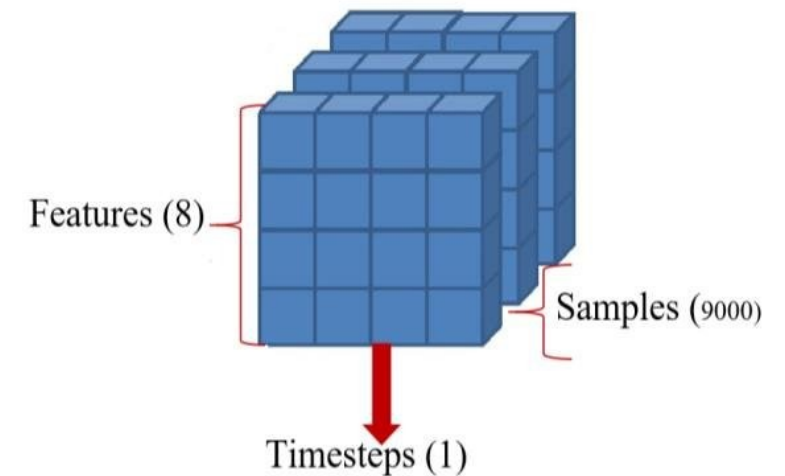
Term-document matrix

Data → Vectors, Matrices, Tensors

□ Time Series



timesteps	features		target
	X_1	X_2	y
timestep 0	5	10	7
timestep 1	7	12	4
timestep 2	3	18	6
timestep 3	8	16	9
timestep 4	2	11	2
timestep 5	9	21	1



Data → Vectors, Matrices, Tensors

□ Audio

```
import wave
import numpy as np
import matplotlib.pyplot as plt

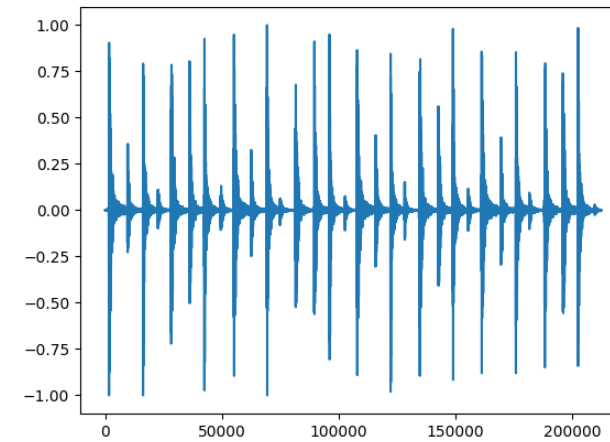
with wave.open("Bongo_sound.wav") as wav_file:
    metadata = wav_file.getparams()
    frames = wav_file.readframes(metadata.nframes)

pcm_samples = np.frombuffer(frames, dtype="<h")
normalized_amplitudes = pcm_samples / (2 ** 15)

plt.plot(normalized_amplitudes);
```

```
print(normalized_amplitudes)
```

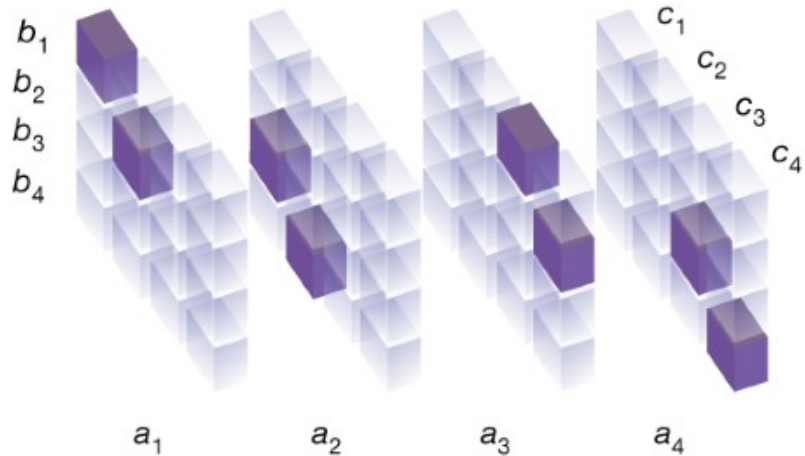
```
[ 3.05175781e-05 -6.10351562e-05  6.10351562e-05 ... -5.49316406e-04
 -5.79833984e-04 -3.96728516e-04]
```



Data → Vectors, Matrices, Tensors

❑ Neural Net Weights

$$\begin{pmatrix} c_1 & c_2 \\ c_3 & c_4 \end{pmatrix} = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \cdot \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix}$$



$$m_1 = (a_1 + a_4)(b_1 + b_4)$$

$$m_2 = (a_3 + a_4)b_1$$

$$m_3 = a_1(b_2 - b_4)$$

$$m_4 = a_4(b_3 - b_1)$$

$$m_5 = (a_1 + a_2)b_4$$

$$m_6 = (a_3 - a_1)(b_1 + b_2)$$

$$m_7 = (a_2 - a_4)(b_3 + b_4)$$

$$c_1 = m_1 + m_4 - m_5 + m_7$$

$$c_2 = m_3 + m_5$$

$$c_3 = m_2 + m_4$$

$$c_4 = m_1 - m_2 + m_3 + m_6$$

$$\mathbf{U} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & -1 \end{pmatrix}$$

$$\mathbf{V} = \begin{pmatrix} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\mathbf{W} = \begin{pmatrix} 1 & 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

NumPy vs Tensorflow

- ❑ **NumPy** provides support for large multidimensional arrays (vectors), matrices and tensors.

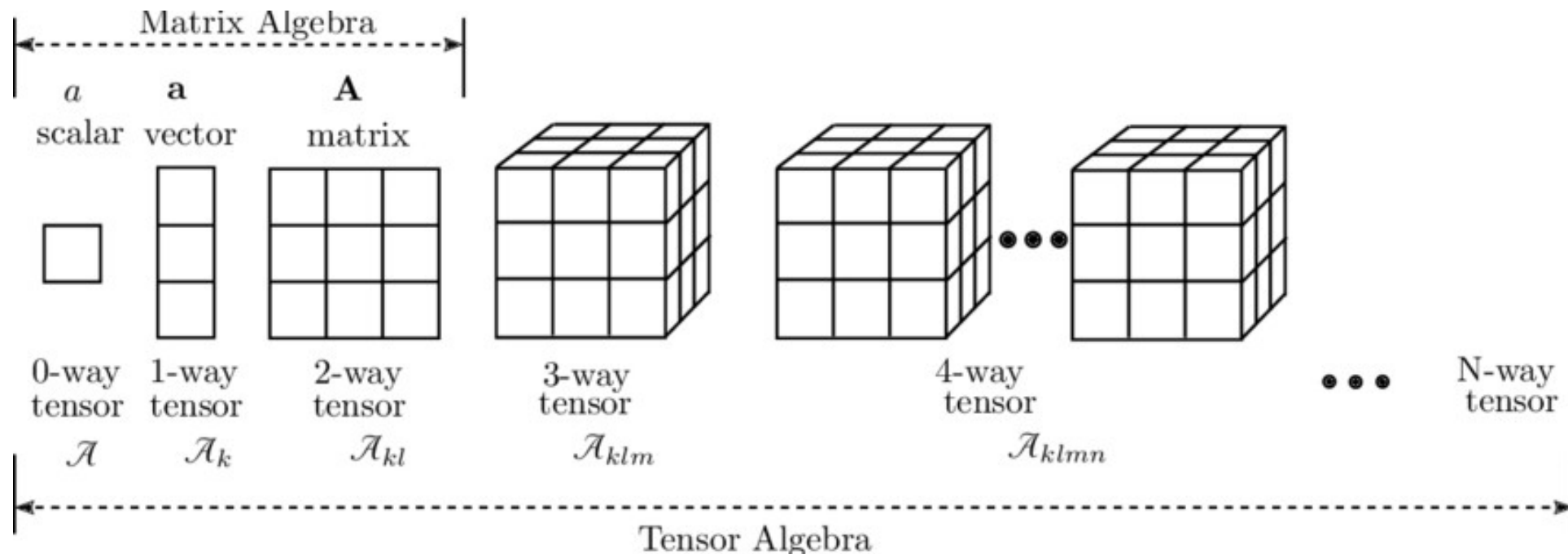


- ❑ **Tensorflow** also provides support for large multidimensional arrays (vectors), matrices and tensors. Has the ability to deploy numerical manipulations to run on CPUs, GPUs.



Linear Algebra

- study of vectors / multidimensional arrays and certain algebra rules to manipulate them.



Vectors

- A vector is an object that has both a magnitude and a direction. In mathematics and physics, vector is a term that refers to some quantities that cannot be expressed by a single number (a scalar).

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{bmatrix}$$

Vectors

- ❑ A vector is an object that has both a magnitude and a direction. In mathematics and physics, vector is a term that refers to some quantities that cannot be expressed by a single number (a scalar).

 NumPy

```
# NumPy
import numpy as np

v1 = np.array([1,2,3,4])
```

```
v1.shape
```

Vectors

- ❑ A vector is an object that has both a magnitude and a direction. In mathematics and physics, vector is a term that refers to some quantities that cannot be expressed by a single number (a scalar)

 TensorFlow

```
# Tensorflow
import tensorflow as tf

v1 = tf.constant([1,2,3,4])
```

```
# this is a 1 dimension tensor
v1.ndim
```

1

```
v1.shape
```

TensorShape([4])



Python

Vector Operations (+, -, scalar mult)

- Adding vectors means adding the individual elements.
- Multiplying vectors by scalars means multiplying each element by scalar

$$\begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \\ 4 \end{pmatrix}$$

```
# vector addition
v2 = np.array([2,3,4,5])

v1 + v2

array([3, 5, 7, 9])

# scalar multiplication
2*v1

array([2, 4, 6, 8])
```

$$k\mathbf{v} = 4 \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \cdot 2 \\ 4 \cdot (-3) \\ 4 \cdot 5 \end{pmatrix} = \begin{pmatrix} 8 \\ -12 \\ 20 \end{pmatrix}$$

Vectors

- ❑ Adding vectors means adding the individual elements.
- ❑ Multiplying vectors by scalars means multiplying each element by scalar

```
# vector addition  
v2 = np.array([2,3,4,5])  
  
v1 + v2
```

```
array([3, 5, 7, 9])
```

```
# scalar multiplication  
2*v1
```

```
array([2, 4, 6, 8])
```

Vectors

- ❑ Adding vectors means adding the individual elements.
- ❑ Multiplying vectors by scalars means multiplying each element by scalar

```
# vector addition  
v2 = tf.constant([2,3,4,5])  
  
v1 + v2
```

```
<tf.Tensor: shape=(4,), dtype=int32, numpy=array([3, 5, 7, 9], dtype=int32)>
```

```
# scalar multiplication  
2*v1
```

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{bmatrix}$$



Python

Vectors Transpose

- The transpose of a vector changes a column vector to a row vector or vice versa.

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_k \end{bmatrix}$$

$$v^T = [v_1 \quad v_2 \quad \dots \quad v_k]$$

Vectors

- ❑ The transpose of a vector changes a column vector to a row vector or vice versa.

```
▶ import numpy as np

## let's create a row vector
v = np.array([[2,3,4]])

print(f'The vector is: {v}')
print(f'The transpose is: \n {v.T}')
```

```
➞ The vector is: [[2 3 4]]
The transpose is:
[[2]
 [3]
 [4]]
```

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_k \end{bmatrix}$$

$$v^T = [v_1 \quad v_2 \quad \dots \quad v_k]$$

Vectors

- ❑ The transpose of a vector changes a column vector to a row vector or vice versa.

```
▶ import tensorflow as tf

## let's create a row vector
v = tf.constant([[2,3,4]])

print(f'The vector is: {v}')
print(f'The transpose is: \n {tf.transpose(v)}')
```

```
➞ The vector is: [[2 3 4]]
The transpose is:
[[2]
 [3]
 [4]]
```

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_k \end{bmatrix}$$

$$v^T = [v_1 \quad v_2 \quad \dots \quad v_k]$$



Python

The Dot Product

- ▣ Very important in ML
- ▣ is the sum of the products of corresponding components.
- ▣ Can be expressed in terms of the transpose: $\mathbf{x}^\top \mathbf{y}$

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

$$\begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \times \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix} = 36$$

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \sum_{i=1}^n u_i v_i$$

The Dot Product

- ▣ Very important in ML
- ▣ is the sum of the products of corresponding components.
- ▣ Can be expressed in terms of the transpose: $\mathbf{x}^T \mathbf{y}$

$$\begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \times \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix} = 36$$

```
# dot product  
v1 = np.array([2,3,4])  
v2 = np.array([6,4,3])  
v1.dot(v2)
```

The Dot Product

- ❑ Very important in ML
- ❑ is the sum of the products of corresponding components.

$$\begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \times \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix} = 36$$

```
# dot product
v1 = tf.constant([2,3,4])
v2 = tf.constant([6,4,3])

tf.tensordot(v1,v2, 1)
```

```
<tf.Tensor: shape=(), dtype=int32, numpy=36>
```




Python

Vector Norms

- Used in ML to regularize (drop insignificant relationships) weights.

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_k \end{bmatrix}$$

Goal is to zero out or diminish weights for insignificant variables

Vector Norms

- ❑ Used in ML to regularize (drop insignificant relationships) weights.

$$w = \begin{bmatrix} 0.44 \\ 2.83 \\ -0.01 \end{bmatrix}$$

We want this weight to be 0 or as close to zero as possible

x_1 = sq. feet

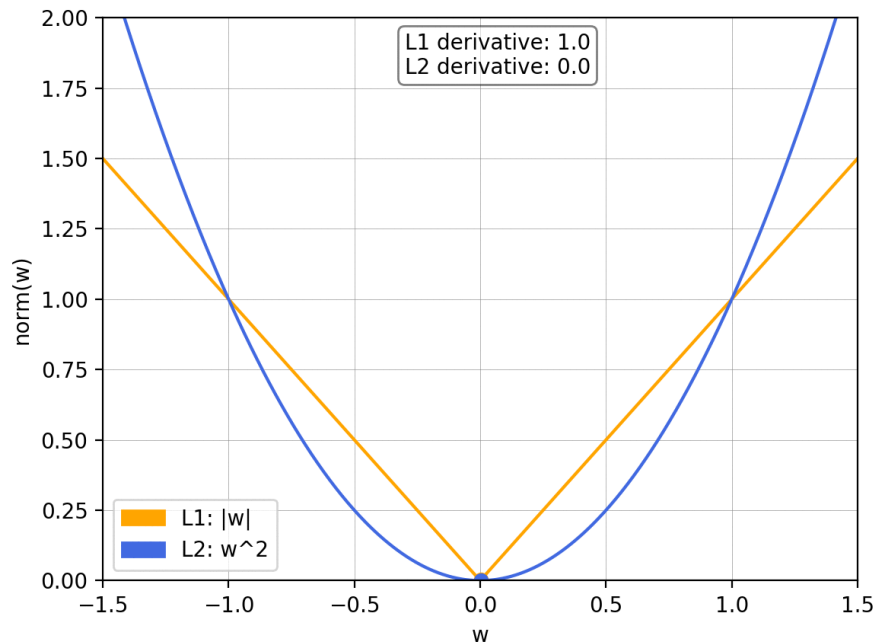
x_2 = # rooms

x_3 = # bulbs

y = Home Price

Vector Norms

- ❑ Are added to cost functions as penalties (e.g., SSE, MSE)
- ❑ The L_1 norm (tends to zero out the unimportant weights).
- ❑ The L_2 norm (tends to get the weights to be close to zero).



$$Cost = SSE + \|\mathbf{w}\|_2$$

$$Cost = SSE + \|\mathbf{w}\|_1$$

L₂ Norm

- ❑ Tends to make weights close to zero
- ❑ For a weight vector:

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_k \end{bmatrix}$$

- ❑ The L2 norm is defined as:

$$\|\mathbf{w}\|_2 = \sqrt{\sum_{i=1}^k w_i^2}$$

$$\|\mathbf{w}\|_2 = \sqrt{\mathbf{w}^T \mathbf{w}}$$

L₂ Norm

$$\mathbf{w} = \begin{bmatrix} 0.44 \\ 2.83 \\ -0.01 \end{bmatrix} \quad \|\mathbf{w}\|_2 = \sqrt{\mathbf{w}^T \mathbf{w}}$$

```
# NumPy
import numpy as np

w = np.array([0.44, 2.83, -0.01])
np.linalg.norm(w, ord = 2)
```

```
➞ 2.8640181563670297
```

L₂ Norm

$$\mathbf{w} = \begin{bmatrix} 0.44 \\ 2.83 \\ -0.01 \end{bmatrix} \quad \|\mathbf{w}\|_2 = \sqrt{\mathbf{w}^T \mathbf{w}}$$

```
[3] # Tensorflow
import tensorflow as tf

w = tf.constant([0.44, 2.83, -0.01])
tf.norm(w, ord = 2)

<tf.Tensor: shape=(), dtype=float32, numpy=2.864018>
```

L₁ Norm

- ❑ Tends to make weights zero
- ❑ For a weight vector:

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_k \end{bmatrix}$$

- ❑ The L1 norm is defined as:

$$\|\mathbf{w}\|_1 = \sum_{i=1}^k |w_i|$$

L₁ Norm



$$\mathbf{w} = \begin{bmatrix} 0.44 \\ 2.83 \\ -0.01 \end{bmatrix} \quad \|\mathbf{w}\|_1 = \sum_{i=1}^k |w_i|$$

```
[4] # NumPy
import numpy as np

w = np.array([0.44, 2.83, -0.01])
np.linalg.norm(w, ord = 1)
```

3.28

L₁ Norm

$$\mathbf{w} = \begin{bmatrix} 0.44 \\ 2.83 \\ -0.01 \end{bmatrix} \quad \|\mathbf{w}\|_1 = \sum_{i=1}^k |w_i|$$

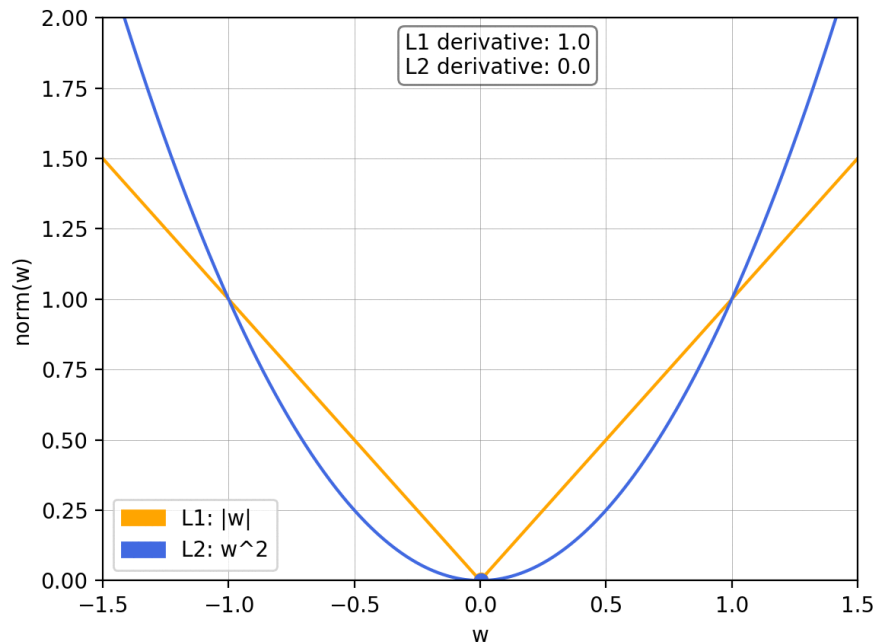
```
[5] # Tensorflow
import tensorflow as tf

w = tf.constant([0.44, 2.83, -0.01])
tf.norm(w, ord = 1)

<tf.Tensor: shape=(), dtype=float32, numpy=3.28>
```

Vector Norms

- Are added to cost functions as penalties (e.g., SSE, MSE)
- The L_1 norm (tends to zero out the unimportant weights).
- The L_2 norm (tends to get the weights to be close to zero).



$$Cost = SSE + \lambda \| \mathbf{w} \|_2$$

$$Cost = SSE + \lambda \| \mathbf{w} \|_1$$

λ is used to control amount of penalty



Python

Matrices

- A matrix is a rectangular array of numbers (called elements) arranged into rows and columns.

$$\begin{array}{c} \text{Columns} \\ \begin{array}{cccc} 1 & 2 & \cdots & n \end{array} \\ \begin{array}{c} \text{Rows} \\ \left\{ \begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{array} \right\} \end{array} \end{array} \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array} \right] = A_{m \times n}$$

Matrices

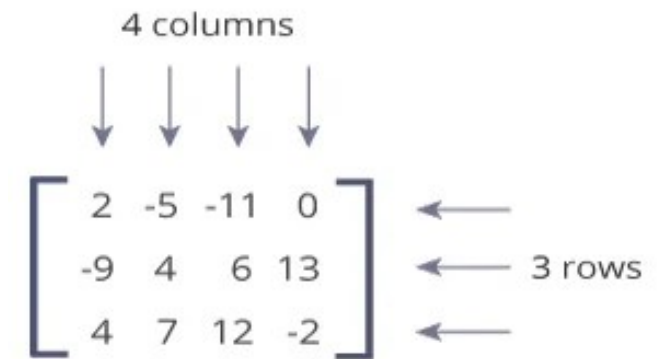
- ❑ A matrix is a rectangular array of numbers (called elements) arranged into rows and columns.

 NumPy

```
M1 = np.array([[2, -5, -11, 0],  
               [-9, 4, 6, 13],  
               [4, 5, 12, -2]])
```

```
M1.shape
```

```
(3, 4)
```



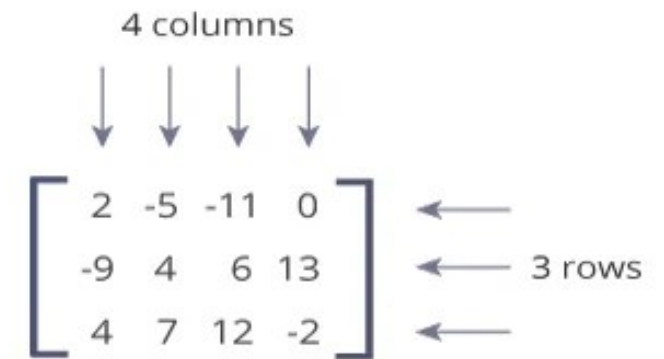
Matrices

- ❑ A matrix is a rectangular array of numbers (called elements) arranged into rows and columns.

 **Tensorflow**

```
M1 = tf.constant([[2, -5, -11, 0],  
                 [-9, 4, 6, 13],  
                 [4, 5, 12, -2]])
```

```
## number of dimensions of the tensor  
M1.ndim
```



Matrix Addition / Subtraction

- ❑ You can add two matrices, if they have the same dimensions.

$$\begin{array}{cc} \text{A} & \text{B} \\ \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix} & + \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{bmatrix} \end{array}$$

Matrix Addition / Subtraction

- ❑ You can add two matrices if they have the same dimensions

$$\begin{matrix} \text{A} & & \text{B} \\ \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix} & + & \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{bmatrix} \end{matrix}$$

 NumPy

```
A = np.array([[4,8],[3,7]])  
B = np.array([[1,0],[5,2]])  
A+B
```

```
array([[5, 8],  
       [8, 9]])
```

Matrix Addition / Subtraction

- ❑ You can add two matrices if they have the same dimensions

$$\begin{matrix} A & & B \\ \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix} & + & \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{bmatrix} \end{matrix}$$

 **Tensorflow**

```
A = tf.constant([[4,8],[3,7]])  
B = tf.constant([[1,0],[5,2]])  
A+B
```

```
<tf.Tensor: shape=(2, 2), dtype=int32, numpy=  
array([[5, 8],  
       [8, 9]], dtype=int32)>
```



Python

Matrix Multiplication

- ❑ A matrix can be multiplied by any other matrix that has the same number of rows as the first has columns (conformable).
- ❑ In other words, Inner Dimensions must match
 - $\mathbf{A}_{m1 \times n1} * \mathbf{B}_{m2 \times n2}$ can be calculated as long as $n1 = m2$
 - $\mathbf{A}_{2 \times 3} * \mathbf{B}_{3 \times 2}$ can be calculated
 - $\mathbf{A}_{3 \times 3} * \mathbf{B}_{3 \times 2}$ can be calculated
 - $\mathbf{A}_{3 \times 2} * \mathbf{B}_{4 \times 2}$ cannot be calculated
 - $\mathbf{A}_{3 \times 2} * \mathbf{B}_{2 \times 7}$ can be calculated
 - $\mathbf{A}_{1 \times 2} * \mathbf{B}_{2 \times 12}$ can be calculated

Matrix Multiplication

- ❑ A matrix can be multiplied by any other matrix that has the same number of rows as the first has columns (conformable).
- ❑ In other words, Inner Dimensions must match
- ❑ Resulting matrix have outer dimensions
 - $\mathbf{A}_{2 \times 3} * \mathbf{B}_{3 \times 2} = \mathbf{C}_{2 \times 2}$
 - $\mathbf{A}_{3 \times 3} * \mathbf{B}_{3 \times 2} = \mathbf{C}_{3 \times 2}$
 - $\mathbf{A}_{3 \times 2} * \mathbf{B}_{2 \times 7} = \mathbf{C}_{3 \times 7}$
 - $\mathbf{A}_{1 \times 2} * \mathbf{B}_{2 \times 12} = \mathbf{C}_{1 \times 12}$
 - $\mathbf{A}_{m1 \times n1} * \mathbf{B}_{n1 \times n2} = \mathbf{C}_{m1 \times n2}$

Matrix Multiplication

- ❑ A matrix can be multiplied by any other matrix that has the same number of rows as the first has columns (conformable).
- ❑ In other words, Inner Dimensions must match
- ❑ Resulting matrix have outer dimensions

$$A_{2 \times 2} = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix} \quad B_{2 \times 2} = \begin{bmatrix} 3 & 6 \\ 7 & 9 \end{bmatrix}$$

$$A * B = C_{2 \times 2} = \begin{bmatrix} 2 * 3 + 1 * 7 = 13 & 2 * 6 + 1 * 9 = 21 \\ 4 * 3 + 5 * 7 = 47 & 4 * 6 + 5 * 9 = 69 \end{bmatrix}$$

Matrix Multiplication



$$A_{2 \times 2} = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix} \quad B_{2 \times 2} = \begin{bmatrix} 3 & 6 \\ 7 & 9 \end{bmatrix}$$

 NumPy

```
import numpy as np

A = np.array([[2,1],[4,5]])
B = np.array([[3,6],[7,9]])

A.dot(B)
```

```
array([[13, 21],
       [47, 69]])
```

```
B.dot(A)
```

```
array([[30, 33],
       [50, 52]])
```

Matrix Multiplication



$$A_{2 \times 2} = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix} \quad B_{2 \times 2} = \begin{bmatrix} 3 & 6 \\ 7 & 9 \end{bmatrix}$$

 TensorFlow

```
import tensorflow as tf
```

```
A = tf.constant([[2,1],[4,5]])
```

```
B = tf.constant([[3,6],[7,9]])
```

```
tf.matmul(A,B)
```

```
<tf.Tensor: shape=(2, 2), dtype=int32, numpy=
array([[13, 21],
       [47, 69]], dtype=int32)>
```

```
tf.matmul(B,A)
```

```
<tf.Tensor: shape=(2, 2), dtype=int32, numpy=
array([[30, 33],
       [50, 52]], dtype=int32)>
```




Python

Matrix Transpose

- Is a matrix that is obtained by interchanging the rows and columns of the matrix

$$\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}^T = \begin{bmatrix} 6 & 1 \\ 4 & -9 \\ 24 & 8 \end{bmatrix}$$

Matrix Transpose



```
import numpy as np

A = np.array([[2,1],[4,5]])
B = np.array([[3,6],[7,9]])
print('-----')
print('Original Matrix A')
print(A)
print('-----')
print("Matrix A'")
print(A.transpose())
```

```
-----
Original Matrix A
[[2 1]
 [4 5]]
-----
Matrix A'
[[2 4]
 [1 5]]
```

Matrix Transpose

```
import tensorflow as tf

A = tf.constant([[2,1],[4,5]])
B = tf.constant([[3,6],[7,9]])
print('-----')
print('Original Matrix A')
print(A)
print('-----')
print("Matrix A'")
print(tf.transpose(A))
```

```
-----
Original Matrix A
tf.Tensor(
[[2 1]
 [4 5]], shape=(2, 2), dtype=int32)
-----
Matrix A'
tf.Tensor(
[[2 4]
 [1 5]], shape=(2, 2), dtype=int32)
```



Python

Identity Matrix

- ❑ Is a matrix of order $n \times n$ such that each main diagonal element is equal to 1

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Identity Matrix



☐ Matrix of order $n \times n$ such that each main diagonal element is equal to 1

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

 NumPy

```
# you can create a square identity matrix with the np.eye() function  
I = np.eye(3)
```

Identity Matrix

☐ Matrix of order $n \times n$ such that each main diagonal element is equal to 1

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

 **Tensorflow**

```
# you can create a square identity matrix with the tf.eye() function  
I = tf.eye(3)  
I
```

```
<tf.Tensor: shape=(3, 3), dtype=float32, numpy=  
array([[1., 0., 0.],  
       [0., 1., 0.],  
       [0., 0., 1.]], dtype=float32)>
```




Python

Linear Dependence

- A matrix is linearly dependent if at least one row (or one column) is a linear combination of the others

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 2 & 4 & 6 \\ 3 & 1 & 9 \end{bmatrix}$$

Linear Dependence

- A matrix is linearly dependent if at least one row (or one column) is a linear combination of the others

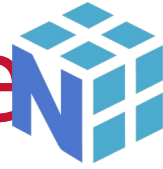
$$A = \begin{bmatrix} 1 & 4 & 3 \\ 2 & 4 & 6 \\ 3 & 1 & 9 \end{bmatrix} \quad 2 \times 2 \quad \begin{matrix} 1 \\ 3 \end{matrix}$$

Linear Dependence

- The rank of a matrix is the number of columns that are independent

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 2 & 4 & 6 \\ 3 & 1 & 9 \end{bmatrix} \quad 2 \times 2$$

Linear Dependence



 NumPy

```
## check the rank  
## if rank < than # of cols or rows then A is dependent  
## note matrix needs to be square
```

```
A = np.array([[1,4,6],  
              [2,4,8],  
              [3,1,7]])  
np.linalg.matrix_rank(A) ## dependent
```

2

```
A = np.array([[2,3],  
              [4,7]])  
np.linalg.matrix_rank(A) ## independent
```

2

Linear Dependence

 TensorFlow

```
## check the rank  
## if rank < than # of cols or rows then A is dependent  
## note matrix needs to be square  
## for tensorflow matrix needs to be float type
```

```
A = tf.constant([[1,4,6],  
                 [2,4,8],  
                 [3,1,7]], dtype = 'float32')  
tf.linalg.matrix_rank(A) ## dependent
```

```
<tf.Tensor: shape=(), dtype=int32, numpy=2>
```

```
A = tf.constant([[2,3],  
                 [4,7]], dtype = 'float32')  
tf.linalg.matrix_rank(A) ## independent
```

```
<tf.Tensor: shape=(), dtype=int32, numpy=2>
```



Python

Matrix Inverse

□ is a matrix A^{-1} such that $AA^{-1} = I$, where I is the identity matrix.

$$\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Matrix Inverse



```
# example with dependent matrix
import numpy as np

A = np.array([[1,4,6],
              [2,4,8],
              [3,1,7]], dtype = 'float32')
np.linalg.det(A) ## dependent

## there is NO inverse
```

0.0

```
A = np.array([[2,3],
              [4,7]], dtype = 'float32')
np.linalg.det(A) ## independent
```

2.0

```
np.linalg.inv(A)
```

```
array([[ 3.5, -1.5],
       [-2. ,  1. ]], dtype=float32)
```

Matrix Inverse

```
# example with dependent matrix
import tensorflow as tf

A = tf.constant([[1,4,6],
                 [2,4,8],
                 [3,1,7]], dtype = 'float32')
tf.linalg.det(A) ## dependent

## there is NO inverse
```

```
<tf.Tensor: shape=(), dtype=float32, numpy=0.0>
```

```
A = tf.constant([[2,3],
                 [4,7]], dtype = 'float32')
tf.linalg.det(A) ## independent
```

```
<tf.Tensor: shape=(), dtype=float32, numpy=2.0>
```

```
tf.linalg.inv(A)
```

```
<tf.Tensor: shape=(2, 2), dtype=float32, numpy=
array([[ 3.5, -1.5],
       [-2. ,  1. ]], dtype=float32)>
```



Python

Sparse vs Dense Matrices

- ❑ a sparse matrix is a matrix in which most of the elements are zero

Dense Matrix

1	2	31	2	9	7	34	22	11	5
11	92	4	3	2	2	3	3	2	1
3	9	13	8	21	17	4	2	1	4
8	32	1	2	34	18	7	78	10	7
9	22	3	9	8	71	12	22	17	3
13	21	21	9	2	47	1	81	21	9
21	12	53	12	91	24	81	8	91	2
61	8	33	82	19	87	16	3	1	55
54	4	78	24	18	11	4	2	99	5
13	22	32	42	9	15	9	22	1	21

Sparse Matrix

1	.	3	.	9	.	3	.	.	.
11	.	4	.	.	.	.	.	2	1
.	.	1	.	.	.	4	.	1	.
8	.	.	.	3	1	.	.	.	.
.	.	.	9	.	.	1	.	17	.
13	21	.	9	2	47	1	81	21	9
.	.	.	.	.	.	.	.	.	.
.	.	.	.	19	8	16	.	.	55
54	4	.	.	.	11	.	.	.	.
.	.	2	.	.	.	.	22	.	21

Sparse vs Dense Matrices

```
## sparse matrix
A = np.array([[1, 0, 0, 1, 0, 0], [0, 0, 2, 0, 0, 1], [0, 0, 0, 2, 0, 0]])
print(A)
```

```
[[1 0 0 1 0 0]
 [0 0 2 0 0 1]
 [0 0 0 2 0 0]]
```

If sparse matrices are big, Python will use a different method of storing for efficiency.

```
import scipy.sparse as sparse

# convert to sparse matrix (CSR method)
S = sparse.csr_matrix(A)
print(S)
```

```
(0, 0)      1
(0, 3)      1
(1, 2)      2
(1, 5)      1
(2, 3)      2
```

Sparse vs Dense Matrices

We can revert a sparse matrix to a normal matrix by using the `.todense()` or `.toarray()` numpy methods.

```
# convert to dense  
S.todense()
```

```
matrix([[1, 0, 0, 1, 0, 0],  
        [0, 0, 2, 0, 0, 1],  
        [0, 0, 0, 2, 0, 0]])
```

```
# convert to dense  
S.toarray()
```

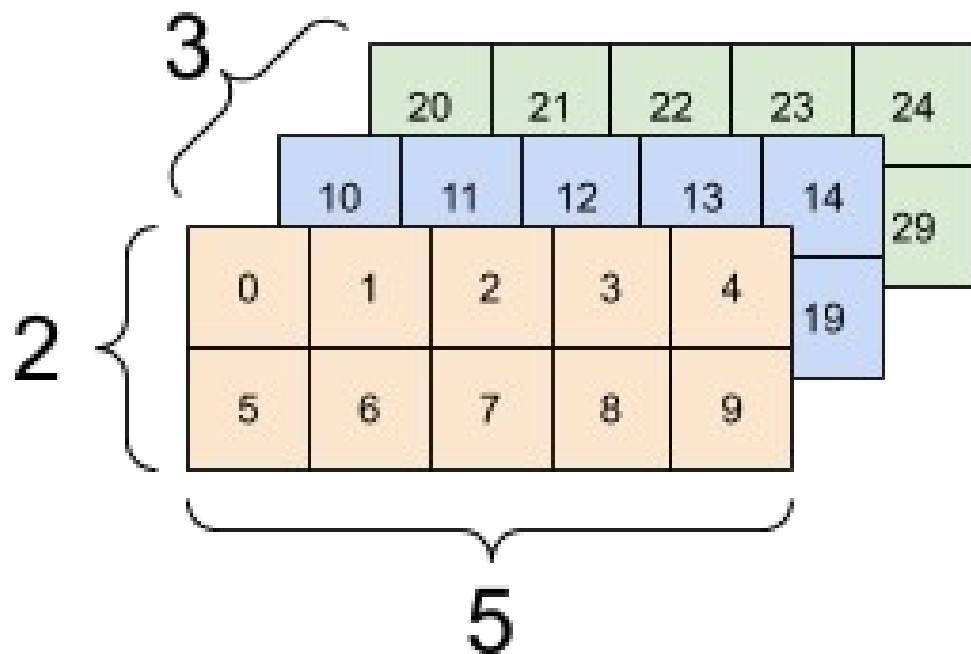
```
array([[1, 0, 0, 1, 0, 0],  
       [0, 0, 2, 0, 0, 1],  
       [0, 0, 0, 2, 0, 0]])
```



Python

Tensors

- A Tensor is a generalization of Vectors and Matrices to higher dimensions



Tensors



 NumPy

```
import numpy as np

# let's create a tensor of 3 matrices, each 3 by 3 (3,3,3)
t1 = np.array([[[10, 11, 12], [13, 14, 15], [16, 17, 18]],
               [[20, 21, 22], [23, 24, 25], [26, 27, 28]],
               [[30, 31, 32], [33, 34, 35], [36, 37, 38]]])
```

t1

```
array([[[10, 11, 12],
        [13, 14, 15],
        [16, 17, 18]],
       [[20, 21, 22],
        [23, 24, 25],
        [26, 27, 28]],
       [[30, 31, 32],
        [33, 34, 35],
        [36, 37, 38]]])
```

Tensors



TensorFlow

```
import tensorflow as tf

# let's create a tensor of 3 matrices, each 3 by 3 (3,3,3)
t1 = tf.constant([[[10, 11, 12], [13, 14, 15], [16, 17, 18]],
                  [[20, 21, 22], [23, 24, 25], [26, 27, 28]],
                  [[30, 31, 32], [33, 34, 35], [36, 37, 38]]])
```

```
t1.shape
```

```
TensorShape([3, 3, 3])
```

```
t1.ndim
```



Python